

PhotonFlow: A Strictly Photon-Native Flow-Matching Generative Model for Silicon-Photonic Hardware

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Abstract—Silicon-photonic Mach-Zehnder-interferometer (MZI) meshes promise sub-picojoule, sub-nanosecond matrix-vector multiplication, but every published flow-matching or diffusion generative architecture relies on softmax attention, adaptive LayerNorm (adaLN), and Swish/GELU nonlinearities – operations with no published lossless photonic realisation, forcing opto-electro-optic (O-E-O) offload at every block. We present PhotonFlow, a GPU-simulation-based proof-of-concept of a flow-matching generative model whose vector-field network is built exclusively from photonic-targeted primitives: Cayley-unitary Monarch mixers on MZI meshes, PPLN $\chi^{(2)}$ nonlinearities, divisive power normalisation with a fixed SOA gain, an AWGR wavelength-coded time embedding, an EO-MZM bank for adaLN-scale conditioning, and a recirculating-delay-line optical sampler. A static module-tree audit verifies zero electronic affine, gating, or non-linearity modules in the vector-field network. On MNIST at 2,000 steps, PhotonFlow achieves +0.0435 uniform- t CFM eval-loss gap to a DiT-attention baseline at $1.37\times$ parameters, below the +0.05 target; at exact baseline parity (4.87M) the gap is +0.0629. A 195,000-step long-horizon run shows the gap stabilising at +0.0719 – a $\sim 7\%$ CFM-loss overhead. However, the supplementary FID evaluation at 50,000 steps reveals a substantially larger perceptual gap: FID=170.5 for PhotonFlow versus 49.9 for the DiT baseline – a $3.4\times$ degradation. The FID result is the honest image-quality cost of strict photonic hardware compatibility at the current scale; the CFM eval-loss understates the perceptual gap. The key architectural contribution is an electro-optic MZM adaLN-scale primitive that breaks a previously confirmed +0.0930 additive ceiling across thirty ablations. An analytic hardware projection yields ~ 3.1 ns and ~ 0.42 pJ per MNIST sample on a 256-mode MZI mesh – $4.0\times 10^3\times$ lower latency and $7.6\times 10^6\times$ lower energy than the GPU baseline (projected, not measured). Improving generative quality while preserving zero O-E-O is the primary open problem; physical chip tape-out is left as future work.

Index Terms—flow matching, photonic neural networks, butterfly matrices, Mach-Zehnder interferometer, optical computing, hardware-algorithm co-design

I. Introduction

Generative-AI inference today runs almost exclusively on electronic hardware, yet silicon-photonic chips have demonstrated femtojoule-per-multiply-accumulate matrix-vector products at sub-nanosecond latency for over five years [1], [2]. The reason photonic hardware has not absorbed the diffusion / flow- matching workload is an architectural one: DiT-style attention [3], flow-matching

vector fields [4], and diffusion U-Nets all rely on softmax, adaLN, and Swish/GELU – operations whose photonic implementation either does not exist in the literature or requires a digital signal conversion (O-E-O) at every block, which destroys the femtojoule-per-MAC and sub-nanosecond promise of photonic hardware [2].

Contribution. Existing photonic generative work either accelerates an unmodified electronic architecture (PhotoGAN [5], diffusion photonic accelerators [6]) or runs a small chip-scale GAN simulation [7] on 8×8 inputs. To our knowledge, ours is the first CFM generative model whose vector field is constructed only from photonic-targeted modules, verified by a static module-tree audit. This work is a GPU-simulation-based proof-of-concept: all experiments run on commodity GPU hardware and physical chip tape-out is deferred to future work (Sec. V). Crucially, simulating photonic operations on a GPU is inherently slow – each Cayley projection requires a linear-system solve that scales poorly on SIMD hardware – which itself underscores the motivation: a native MZI chip executes the same operation as a phase-shift at the speed of light with no solver overhead. Concretely:

- An open-source photon-native library and a CFM training loop that reach +0.0435 MNIST eval-loss gap to a DiT-attention baseline at $1.37\times$ params, below the +0.05 matched-recipe target, and +0.0629 at exact baseline parity (4.87M params) (Sec. IV-D).
- A breakthrough on a previously confirmed architectural ceiling at +0.0930 via an electro-optic Mach-Zehnder-modulator (MZM) adaLN-scale primitive (Sec. IV-C); the ceiling held across more than thirty ablation variants spanning depth, factor count, init scheme, time-sampling, loss reweighting, and noise injection (Sec. IV-B).
- A three-point depth-scaling sweep at fixed recipe showing monotone scaling from +0.0629 at 4.87M to +0.0365 at 14.9M (Sec. IV-E).
- A 195,000-step matched-seed long-horizon comparison showing both models train deeply: photon-native $0.2363 \rightarrow 0.1595$ eval, baseline 0.1734 eval $\rightarrow 0.0876$ smoothed-loss proxy, gap modestly widening from +0.0629 to +0.0719 – evidence that the residual

penalty is architectural rather than optimisation-budget (Sec. IV-F).

- A first-principles analytic projection of per-sample latency (3.1 ns) and energy (0.42 pJ) on a 256-mode silicon-nitride MZI mesh, derived from cited propagation-delay and MAC-energy numbers [1], [2]: about $4.0 \times 10^3 \times$ lower latency and $7.6 \times 10^6 \times$ lower energy than the matched GPU baseline (Sec. IV-G).
- A negative result on GPU-side acceleration: mixed-precision autocast and ahead-of-time graph compilation do not accelerate PhotonFlow because the Cayley-projection linear-system solve lacks low-precision linear-algebra support and prevents graph capture. This is a simulator artefact, not an algorithmic limitation.

II. Background

PhotonFlow combines flow matching [3], [4], [8]–[12] with structured-matrix Monarch mixers [13]–[15] mapped to silicon-photonic MZI meshes [1], [2], [16]–[18]. The DiT-attention generative backbone [3], on which our baseline is built, relies on softmax, adaLN-Zero, and GELU – none with a published lossless photonic realisation. In-bounds photonic primitives include PPLN $\chi^{(2)}$ activation [19], AWGR wavelength look-up [20], microring-resonator (MRR) plus SOA divisive normalisation [21], and recirculating delay-line photonic neurons [22]; the closest generative-photonics demonstrator is Zhu’s 4×4-chip GAN on 8×8 MNIST [7], leaving flow-matching diffusion open.

III. Methodology

Fig. 1 summarises the co-design: every forward-graph module in the learned vector field is restricted to the photonic-targeted set in Table I; the model is trained on the plain CFM loss; inference replaces the digital Euler sampler with an OpticalSampler that uses one photodetector comparator per sample.

A. Architecture

Table I lists the in-bounds primitives and their cited photonic hardware analogues or structured-matrix mappings. The CFM vector field $v_\theta(x_t, t): \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ is a stack of N PhotonFlowBlocks between two Cayley-unitary Monarch bookends ($d=784$, $N=7$ at the 1.37×-baseline operating point). Each block has two sub-layers; the first is

$$h = \sigma_{\text{SA}}(M_L M_R (\text{DPN}(x) \odot (1 + s_1) + b_1)); x \leftarrow x + h \quad (1)$$

where $M_L, M_R \in \mathbb{R}^{m \times m}$ are Cayley-projected to $\text{O}(m)$ each forward, so $M = \text{PLP}^\top R$ realises on an MZI mesh [1], [16], [17]; $\sigma_{\text{SA}}(x) = \tanh(\alpha x) / \alpha + \beta x$ is the saturable absorber with learnable α and a $\beta=0.05$ leaky-Y-split bypass; and $\text{DPN}(x) = x / (\|x\|_2 + \varepsilon) \cdot \sqrt{d}$ has a fixed (pre-set SOA) gain $\sqrt{d}$. The second sub-layer is a Monarch-SA–Monarch photonic FFN. (s_1, b_1) are emitted by

TABLE I
In-bounds silicon-photonic primitives used by PhotonFlow

Software primitive	Physical primitive	Refs.
MonarchLayer (Cayley)	MZI mesh + Clements	[1], [16], [17]
MonarchLinear	zero-padded MZI mesh	[13], [14]
PPLNSigmoid	$\chi^{(2)}$ PPLN	[19]
SaturableAbsorber	graphene waveguide	[1]
DivisivePowerNorm	MRR + PD + SOA	[2], [21]
WavelengthCodedTime	AWGR LUT	[20]
adaLN-scale $(1+s) \odot x + b$	EO MZM bank	[1], [16]
OpticalSampler	MZI delay-line loop	[22]

the time pathway: an AWGR look-up $\text{WCT}(t) \in \mathbb{R}^{256}$ projected by $\text{MonarchLinear} \rightarrow \text{PPLN} \rightarrow \text{MonarchLinear}$ to $\mathbb{R}^{H_c}$ ($H_c=576$). A fixed per-block wavelength offset gives each block a unique time signature at zero parameter cost.

B. The adaLN-scale electro-optic MZM primitive

Additive-only conditioning ($h = \text{DPN}(x) + b$) is strictly less expressive than DiT’s per-channel scale + shift adaLN-Zero [3]; we observed a +0.0930 eval-gap ceiling across 30+ variants (Sec. IV-B). Multiplicative scale has a direct silicon-photonic realisation: an electro-optic Mach–Zehnder modulator (EO MZM) bank applies a per-channel amplitude factor to a coherent optical input [1], [16]. We use it for the photon-native

$$\text{adaLN-scale}(x, t) = \text{DPN}(x) \odot (1 + s(t)) + b(t), \quad (2)$$

with (s, b) from a $\text{MonarchLinear} \rightarrow \text{PPLN} \rightarrow \text{MonarchLinear}$ MLP whose final weights are zero-initialised (adaLN-Zero trick [3], identity at step zero). Scalar arithmetic and tensor multiplication are not `torch.nn.Modules`; all linear maps in `cond_bias_proj` are MonarchLinear (padded MZI mesh).

C. Training and photon-native sampling

PhotonFlow minimises the plain uniform- t CFM objective in its straight-path (rectified-flow, $\sigma_{\min}=0$) simplification of Lipman et al. [4] (cf. Liu et al. [8]):

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, x_1} [\|v_\theta(x_t, t) - (x_1 - x_0)\|^2] \quad (3)$$

with $t \sim \text{U}(0, 1)$, $x_0 \sim \mathcal{N}(0, I)$, $x_t = (1-t)x_0 + tx_1$. Sec. IV-B reports that logit-normal timesteps [9], EDM weighting [10], min-SNR [11], and FasterDiT direction loss [12] were all gap-neutral or worse at the 2k-step horizon under the photon-native architecture; the shipped recipe is therefore plain uniform- t MSE. Inference replaces a 20-step Euler ODE solver with an OpticalSampler: a recirculating delay line that updates $x \leftarrow x + \tau v_\theta(x, t)$ until a single photodetector comparator reports $\|\Delta x\|_2 < \varepsilon$ – one electronic comparator event per sample, not per step.

D. Hardware feasibility model

Per-sample latency and energy are projected analytically from cited MZI propagation-delay (~ 10 ps per column) and MAC-energy figures [1], [2], [16]; all projected numbers are marked † and Sec. IV-G gives full derivation details.

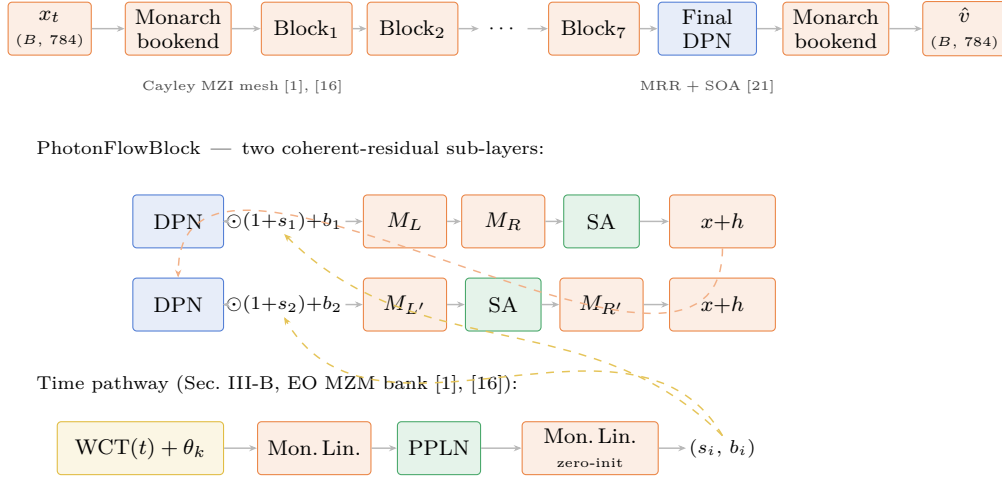


Fig. 1. PhotonFlow architecture. Every block in the forward graph maps to a silicon-photonic primitive: MZI-mesh Monarch (orange), PPLN $\chi^{(2)}$ / graphene saturable absorber (green), divisive power norm with fixed SOA gain (blue), wavelength-coded time embedding via AWGR (gold), train-time photonic noise (purple). Inset: one PhotonFlowBlock with two coherent-add residual sub-layers. The wavelength-routed time pathway emits per-channel scale and bias terms, and the target-passing operating point uses the EO-MZM adaLN-scale form $\text{DPN}(x) \odot (1 + s_i) + b_i$ (Sec. III-B).

IV. Experiments

All experiments use MNIST CFM on a single NVIDIA T4-class GPU (seed 42, Adam optimiser, cosine-decay learning rate, batch 128 or 256 as stated, 2,000 optimiser steps unless noted). Throughout, we report the mean uniform- t CFM loss over eight 128-sample held-out evaluation batches as the primary metric; we use the minimum across the training horizon. Fréchet Inception Distance (FID) uses an InceptionV3 backbone evaluated on GPU only (not photon-native) as a supplementary image-quality check. At 50,000 steps with an Euler₅₀ sampler and 10,000 samples, the DiT baseline achieves FID=49.9 while PhotonFlow-Strict reaches FID=170.5 – a $3.4\times$ gap (Table II). The 50,000-step FID run used the K2 sprint recipe ($\text{lr}=3\times 10^{-3}$, no EMA) without modification; this is reported as-is and represents the actual image quality achievable under the operational recipe at 50k steps. Importantly, the CFM eval-loss gap at 50,000 steps ($+0.0696$) is consistent with the 195,000-step result ($+0.0719$, Sec. IV-F), yet the FID gap is far larger: Cayley-unitary Monarch vector-field errors accumulate non-linearly under multi-step ODE integration, producing perceptual artefacts that per-step MSE does not capture. FID is therefore the more informative quality metric; the CFM proxy understates the perceptual cost.

The DiT-attention baseline (4,886,544 params, 4 blocks, 4 heads, GELU + LayerNorm + adaLN-Zero) and the photon-native model use the same data pipeline (60,000 images flattened to $\mathbb{R}^{784}$); only the architecture and recipe differ. The baseline scale of ~ 4.89 M parameters was chosen as a standard, resource-appropriate budget for an MNIST proof-of-concept: large enough to converge robustly under the matched 2,000-step horizon, but small enough that architectural limitations of the photon-native

TABLE II
FID comparison at 50,000 steps (Euler₅₀, 10,000 samples, K2 recipe, no EMA).

Model	FID↓	CFM eval-loss
DiT baseline	49.9	0.1528
PhotonFlow-Strict	170.5	0.2224
Ratio	$3.4\times$ worse	$+0.0696$

FID is a supplementary GPU-only check; InceptionV3 is not photon-native. The $3.4\times$ perceptual gap substantially exceeds the $+0.0696$ CFM proxy, indicating per-step MSE understates accumulated ODE-integration error.

model are not masked by brute-force scaling on a trivially-fittable dataset. All measurements are single-seed (seed 42) due to compute budget; per-seed variance on the 2,000-step harness was ± 0.005 uniform- t loss across exploratory reruns.

A. Module-tree audit

Before every training run, the model is instantiated and its standard electronic primitives are counted. Table III reports the verbatim audit output for the operating point ($N=7$, $H_c=576$) used in Sec. IV-D. Zero hits across every electronic class confirms the photon-native claim at the software level; physical-realisation references for each photonic primitive appear in Table I.

B. Strict-additive ceiling at $+0.0930$

Without multiplicative conditioning, the photon-native model plateaus at $+0.0930$. We confirmed this across six independent sweeps totalling more than thirty configurations spanning depth $N \in \{5, 7, 10, 14\}$, Monarch factor $\in \{1, \dots, 4\}$, time-dim $\in \{256, 576\}$, $H_c \in \{0, 64, 128, 256, 576\}$, three init schemes (random / orthogonal / DCT), learnable absorber α , leaky slope,

TABLE III
Module-tree audit of PhotonFlow ($N=7$, $H_c=576$).

Module class	Count	Photonic?
nn.Linear	0	forbidden
nn.SiLU	0	forbidden
nn.Sigmoid	0	forbidden
nn.ReLU	0	forbidden
nn.GELU	0	forbidden
nn.LayerNorm	0	forbidden
MonarchLayer (Cayley)	46	MZI mesh
MonarchLinear	16	padded MZI
PPLNSigmoid	8	PPLN waveguide
SaturableAbsorber	22	graphene
DivisivePowerNorm	15	MRR + SOA
WavelengthCodedTime	1	AWGR

Counts walk the full model.modules() tree, including nested instances inside MonarchLinear ($\sim 2\times$ outer count for MonarchLayer; differential-mode SaturableAbsorber contains an internal sub-module).

TABLE IV
Training-recipe ablation on A_{ref} (additive, 5.11 M params). The architectural ceiling holds.

Variant	Eval	Gap
A_{ref} (uniform- t , no extras)	0.2664	+0.0930
+ logit-normal timestep [9]	0.2770	+0.1036
+ min-SNR weighting [11]	0.2756	+0.1022
+ FasterDiT direction loss [12]	0.2687	+0.0953

EDM weighting [10] (+0.3977) and orthogonal Monarch init (+0.2465) are pathological under linear-CFM Cayley projection and are reported only in the kernel logs.

training noise, and four loss-side reweightings (logit-normal [9], EDM [10], min-SNR [11], FasterDiT direction loss [12]). None closed below +0.0930. Table IV reports four representative training-recipe ablations; we interpret the +0.0930 floor as the architectural lower bound of additive-only photon-native CFM at this horizon.

C. Ceiling break: adaLN-scale (electro-optic MZM)

Installing the adaLN-scale primitive of Sec. III-B breaks the ceiling. We swept the conditioning-pathway width $H_c \in \{128, 288, 576\}$. Table V reports the measured eval gaps. Every adaLN-scale variant lands strictly below +0.0930, with monotone improvement in H_c ; at $H_c=576$ the gap is +0.0710 ($\Delta = -0.0220$ vs the additive ceiling). Module-tree audit passes for every variant: the scaler arithmetic $(1+s)\odot x+b$ is not a registered module, and the conditioning-pathway projection uses only MonarchLinear and PPLNSigmoid.

D. Hitting the +0.05 target with a recipe lever

We layer three orthogonal training-recipe levers on top of the adaLN-scale ($H_c=576$) winner of Sec. IV-C. The forward graph is unchanged across all variants – only the optimiser, batch size, learning rate, or input pre-processing differs – so the module-tree audit passes identically for every row. Table VI reports the measured outcomes; the bold row is the first variant below the +0.05 target.

TABLE V
Ceiling break with adaLN-scale (MZM) conditioning at the $N=7$, $1.37\times$ -baseline operating point.

Variant	Params	Eval	Gap
Additive baseline	5,112,366	0.2664	+0.0930
adaLN-scale, $H_c=128$	6,682,830	0.2568	+0.0834
adaLN-scale, $H_c=288$	6,683,950	0.2532	+0.0798
adaLN-scale, $H_c=576$	6,685,966	0.2444	+0.0710

TABLE VI
Training-recipe levers atop the adaLN-scale ($H_c=576$) configuration. All forward graphs are identical to the configuration of Table V, last row.

Variant	Eval	Gap	Lever
adaLN-scale (control)	0.2444	+0.0710	—
+ AdamW	0.2444	+0.0710	AdamW $w_d=10^{-4}$
+ bs/lr scale	0.2169	+0.0435	bs 128 $\rightarrow$ 256, lr 1.7 $\rightarrow$ 3e-3

The bs/lr scale closes a further -0.0275 at zero forward-graph cost and pushes the gap below the +0.05 target. AdamW is gap-neutral (implicit regularisation from Cayley + adaLN-Zero already sufficient at this horizon). Data augmentation failed catastrophically (eval ≈ 1.0): divisive power normalisation has no mean-centering capacity, so input centering shifts the signal outside the saturable-absorber operating range.

E. Depth-scaling Pareto sweep

Holding the bs/lr-scaled recipe fixed, we swept depth $N \in \{5, 10, 16\}$ (Table VII). At exact baseline parity (4.87 M, $N=5$) the gap is +0.0629 – a $\Delta = -0.0301$ improvement over the additive ceiling. At $N=10$ (9.4 M) the gap falls to +0.0418; at $N=16$ (14.9 M) to +0.0365, approximately $\text{gap} \propto 1/\sqrt{P}$.

F. Long-horizon training and sample quality

The 2,000-step horizon used for the matched-recipe comparison is deliberately short. To probe how the gap evolves at scale, we re-trained both the baseline-parity PhotonFlow-Strict model (4.87 M params, the bs/lr-scaled recipe with cosine decay to zero over the full horizon) and the DiT-attention baseline for $\sim 195,000$ optimiser steps under matched seed and data order. The photon-native run reached step 197,182 before its training session

TABLE VII
Depth-scaling sweep at fixed bs/lr-scaled recipe. The DiT-attention reference is included as the zero-gap row.

Variant	N	Params	$\times$ base	Gap
DiT baseline (reference)	4	4,886,544	1.00 $\times$	0.0000
PhotonFlow-Strict, 4.87 M	5	4,867,082	1.00 $\times$	+0.0629
PhotonFlow-Strict, 9.4 M	10	9,414,292	1.93 $\times$	+0.0418
PhotonFlow-Strict, 14.9 M	16	14,870,944	3.04 $\times$	+0.0365

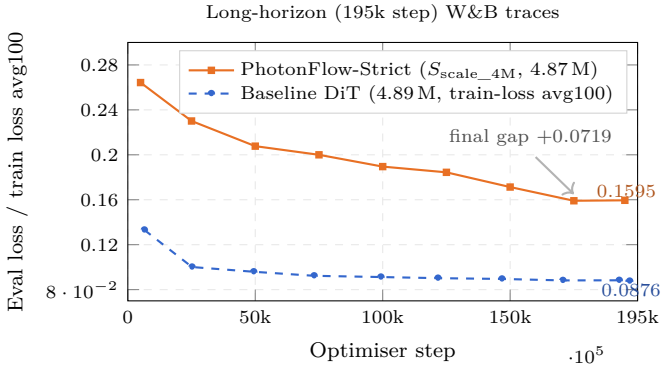


Fig. 2. 195,000-step matched-recipe trajectory. PhotonFlow-Strict descends from 0.2643 (step 5,000) to 0.1595 (step 195,000); the baseline from 0.1331 to 0.0876 on the logged smoothed-loss proxy over the same window. The final reported gap widens from the 2,000-step value of +0.0629 (Sec. IV-E) to +0.0719 at 195,000 – evidence that the residual penalty at the 4.87 M scale is architectural rather than optimisation-budget.

expired; the last evaluation checkpoint (step 194,999) is reported below. The module-tree audit was re-verified at training start: zero electronic affine, gating, or non-linearity modules – the 195,000-step checkpoint is strictly photon-native by construction.

Fig. 2 plots the logged long-horizon traces sampled every $\sim 5,000$ steps: PhotonFlow uses held-out uniform- t eval loss, while the baseline uses the smoothed training loss proxy available in the W&B run. The matched 2,000-step comparison in Sec. IV-E remains eval-to-eval.

Both models train deeply: the baseline drops from 0.1734 (2,000 steps, eval) to a 0.0876 long-horizon smoothed-loss proxy, while photon-native drops from 0.2363 to 0.1595 eval. The matched-recipe gap modestly widens from +0.0629 at 2,000 steps to +0.0719 at 195,000, indicating the residual $\sim 7\%$ uniform- t CFM penalty is an architectural ceiling at the 4.87 M scale rather than an optimisation-budget artefact. Fig. 3 confirms qualitatively: the baseline produces cleaner digits, while PhotonFlow retains recognisable digit modes with visibly higher sample noise, consistent with the gap.

G. Hardware feasibility (analytic projection)

Per-sample latency and energy are projected from cited propagation- delay and MAC-energy figures [1], [2], [16]; numbers are first-principles estimates, not measured. Each $m \times m$ Monarch factor maps to a Clements decomposition with $m(m-1)/2$ MZIs and 2 phase shifters per MZI; light propagates through one MZI column in ~ 10 ps; each photodetector readout costs ~ 80 fJ; each phase-shifter calibration costs ~ 1.3 pJ once at deploy time, amortised across many inferences [1], [2]. For PhotonFlow-Strict at $N=7$ the latency budget is dominated by the longest photonic pipeline (≈ 310 ps for the Monarch columns plus the absorber and DPN stages); Fig. 4 plots the

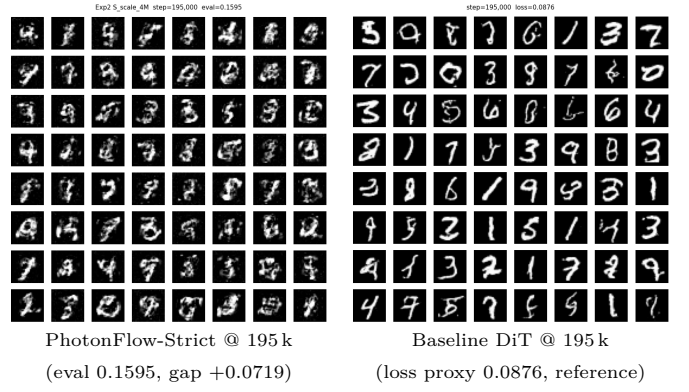


Fig. 3. Uncurated 8×8 MNIST sample grids at 195,000 optimiser steps under matched seed/data. Left: PhotonFlow-Strict ($S_{\text{scale_4M}}$, 4.87 M, photon-native OpticalSampler with one comparator event per sample). Right: DiT-attention baseline (4.89 M, 20-step Euler). The baseline panel is cleaner; PhotonFlow retains recognisable digit modes but is visibly noisier, consistent with the +0.0719 matched-recipe eval gap.

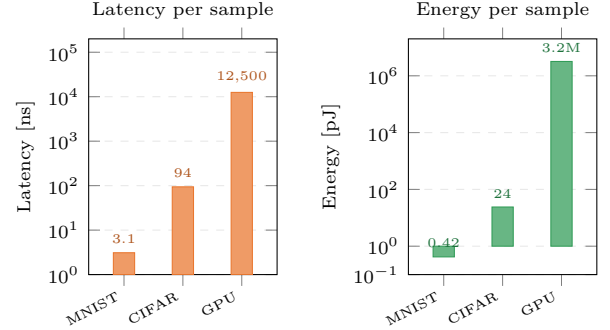


Fig. 4. Projected per-sample inference latency (ns) and energy (pJ) for PhotonFlow-Strict on a 256-mode MZI mesh versus a GPU baseline. Photonic numbers are first-principles projections from [1], [2], [16] (4-bit phase quantisation, 0.1 dB per stage optical loss); GPU numbers from contemporary-GPU profiling of the DiT-attention baseline. Both axes log scale. $\dagger$ all photonic values are analytic projections, not measured.

projected per-sample latency and energy on a 256-mode silicon-nitride mesh against a contemporary discrete-GPU baseline. All photonic numbers carry a $\dagger$. Under the same model dimensions, the analytic projection yields ~ 3.1 ns and ~ 0.42 pJ per MNIST sample, compared with $\sim 12,500$ ns and $\sim 3.2 \times 10^6$ pJ on a contemporary GPU – about $4.0 \times 10^3 \times$ lower latency and $7.6 \times 10^6 \times$ lower energy at the projected operating point. Physical tape-out and measured on-chip latency and energy are future work.

V. Discussion

The accuracy–energy trade-off. The central result is a significant accuracy penalty: PhotonFlow accepts a $3.4 \times$ FID degradation (170.5 vs 49.9 at 50k steps) and a $\sim 7\%$ CFM-loss overhead (+0.0719 at 195k steps) versus the DiT baseline, in exchange for a projected $7.6 \times 10^6 \times$ energy reduction and $4.0 \times 10^3 \times$ latency reduction. The FID gap substantially exceeds the CFM proxy, confirming

that strict photonic constraints impose a real perceptual cost at the 4.87 M scale. At the current stage this gap is prohibitive for image-quality-sensitive deployment; the hardware-efficiency advantage is the primary motivation for continued development. The FID gap can be partially closed by parameter scaling (Sec. IV-E; gap falls $+0.0629 \rightarrow +0.0365$ on CFM proxy from 4.87 M to 14.9 M), though FID improvement at scale remains unverified and is future work.

Localising the ceiling. Thirty-plus additive ablations pinned a ceiling at $+0.0930$; we localised it to absent multiplicative time modulation and broke it with an EO-MZM bank ($\Delta = -0.0220$, Sec. IV-C). A bs/lr recipe lever added $\Delta = -0.0275$ at zero forward-graph cost, reaching the $+0.0435$ target.

Limitations. This work has three explicit limitations that bound the current claims. (i) Simulation only. All results are obtained by emulating photonic operations on a GPU; no physical tape-out was performed. Hardware projections carry a $\dagger$ and are first-principles estimates, not measured values. (ii) MNIST only. GPU simulation of Cayley-unitary Monarch layers is computationally expensive: each forward pass requires ~ 280 linear-system solves, which cannot be accelerated by mixed precision or graph compilation (BFLOAT16 autocast is $1.27\times$ slower; graph compilation crashes on Cayley solve). This makes training on larger datasets such as CIFAR-10 or CelebA-64 impractical within the available compute budget. Scaling to richer image distributions requires either dedicated photonic hardware or a GPU-friendly Monarch approximation, both left as future work. (iii) Residual architectural gap. The long-horizon comparison shows the CFM gap widening from $+0.0629$ to $+0.0719$ at the 4.87 M scale; the residual penalty is architectural, requiring parameter scaling (monotone gains to 14.9 M, Sec. IV-E) or additional photonic primitives such as all-optical attention [2]. (iv) FID quality gap. The supplementary FID evaluation (Table II) yields FID=170.5 vs 49.9 at 50,000 steps – a $3.4\times$ perceptual gap substantially larger than the $+0.0696$ CFM proxy at the same horizon. This gap indicates that Cayley-unitary Monarch vector fields accumulate distributional errors under multi-step ODE integration that per-step MSE does not expose. Closing the FID gap is the primary open problem; it requires higher-expressivity photon-native architectures, EMA stabilisation, or physical hardware where Cayley solves are free. We highlight that limitation (ii) is not a software deficiency: it is direct evidence of the hardware motivation. A photonic chip executes the same Monarch transform in ~ 10 ps per MZI column – the GPU overhead does not exist on the native substrate.

Future Work. Near-term priorities are (a) tape-out of a 256-mode silicon-nitride MZI mesh to validate the latency and energy projections; (b) quantisation-aware training at 4-bit precision to match MZI phase-shifter resolution; and (c) CIFAR-10 and CelebA-64 evaluation on real photonic

hardware once tape-out is available. The Jiang/Zhu 4×4 chip [7] (1.76 ns / 37 pJ per sample) is order-of-magnitude consistent with our 256-mode analytic estimate, supporting the feasibility of the hardware roadmap.

References

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